

Task 5: Session 5 (AIML) - Assignment Questions

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1. String Indexing

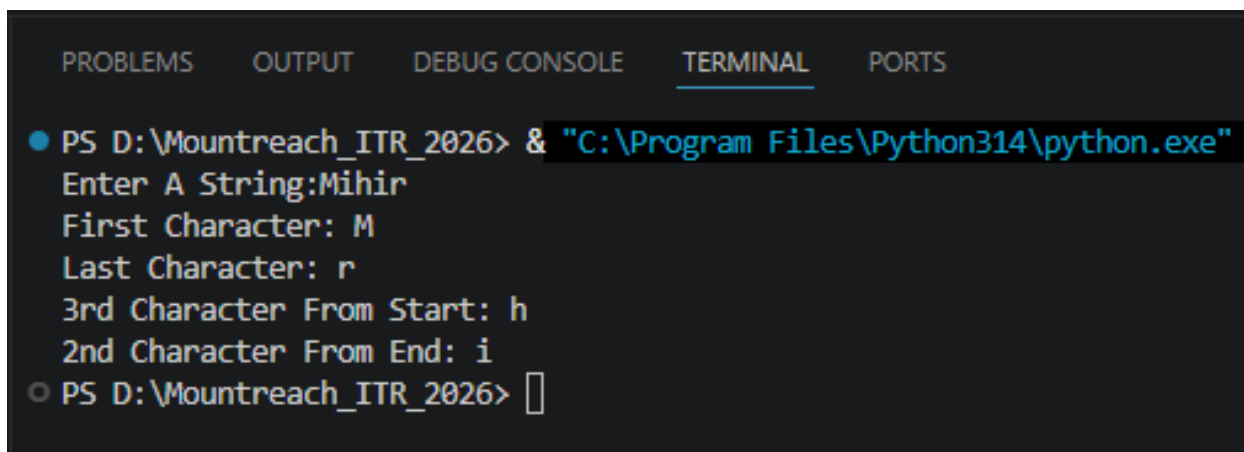
Write a program that takes a string input from the user.

Print:

- The first character
- The last character (using negative index)
- The 3rd character from the start
- The 2nd character from the end

Add comments explaining positive and negative indexing.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
First Character: M
Last Character: r
3rd Character From Start: h
2nd Character From End: i
○ PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
user_string = input("Enter A String:")

# Positive indexing: Starts At 0 From The Left (0, 1, 2, ...)
print("First Character:", user_string[0])
```

```
# Negative indexing: Starts At -1 From The Right (-1, -2, -3, ...)
print("Last Character:", user_string[-1])

# Index 2 Gives The 3rd Character From The Start
print("3rd Character From Start:", user_string[2])

# Index -2 Gives The 2nd Character From The End
print("2nd Character From End:", user_string[-2])
```

2. String Slicing

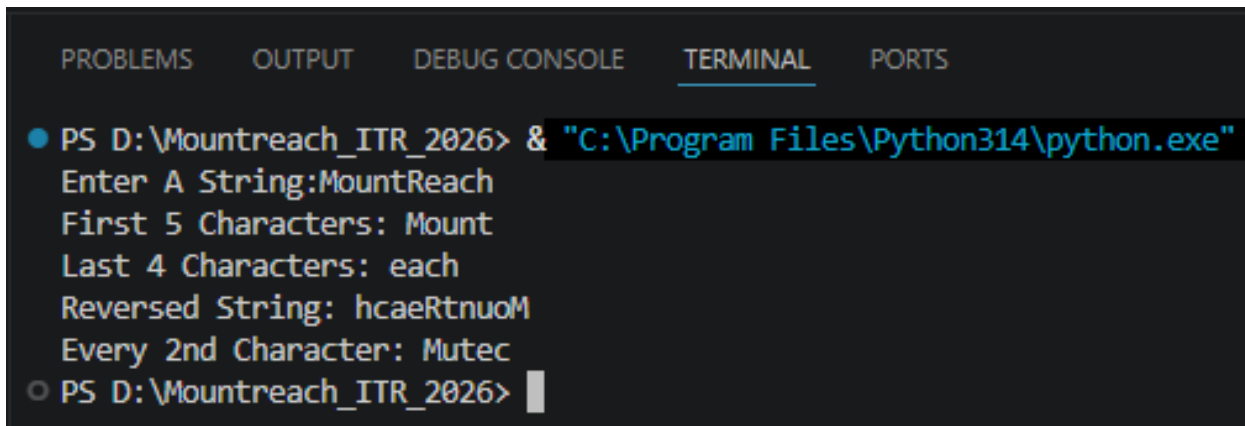
Take a string input from the user.

Using slicing, print:

- The first 5 characters
- The last 4 characters
- The string in reverse order
- Every 2nd character of the string

Include comments explaining the slicing syntax [start:end:step].

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:MountReach
First 5 Characters: Mount
Last 4 Characters: each
Reversed String: hcaeRtnuoM
Every 2nd Character: Mutec
PS D:\Mountreach_ITR_2026>
```

Program Code:

```
user_string = input("Enter A String:")

# Slicing Syntax: string[start:end:step]
# start -> Index To Begin (Inclusive), Default 0
# end   -> Index To Stop (Exclusive), Default len(string)
# step  -> How Many Positions To Jump, Default 1

# start=0 (default), end=5
print("First 5 Characters:", user_string[:5])

# start=-4 (4th From end),
print("Last 4 Characters:", user_string[-4:])

# step=-1 Traverses From Right To Left
print("Reversed String:", user_string[::-1])
```

```
# step=2 Skips One Character Each Time  
print("Every 2nd Character:", user_string[::2])
```

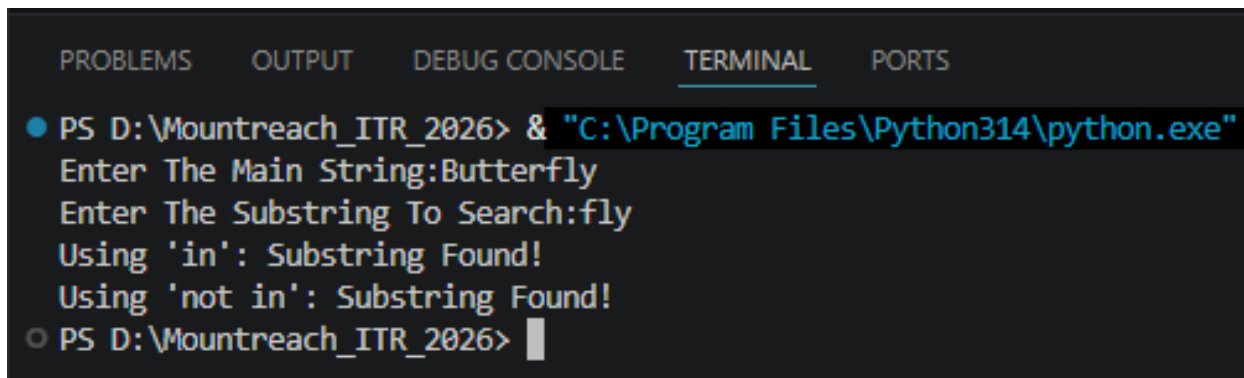
3. String Membership

Write a program that:

- Takes a main string and a substring as input from the user.
- Checks whether the substring is present in the main string using `in` and `not in`.
- Prints appropriate messages (e.g., "Substring found!" or "Substring not found!").

Note that it is case-sensitive.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter The Main String:Butterfly
Enter The Substring To Search:fly
Using 'in': Substring Found!
Using 'not in': Substring Found!
○ PS D:\Mountreach_ITR_2026>
```

Program Code:

```
main_string = input("Enter The Main String:")
sub_string = input("Enter The Substring To Search:")

# 'in' Returns True If sub_string Exists Inside main_string (Case-Sensitive)
if sub_string in main_string:
    print("Using 'in': Substring Found!")
else:
    print("Using 'in': Substring Not Found!")

# 'not in' Returns True If sub_string Does Not Exist Inside main_string
if sub_string not in main_string:
    print("Using 'not in': Substring Not Found!")
else:
    print("Using 'not in': Substring Found!")
```

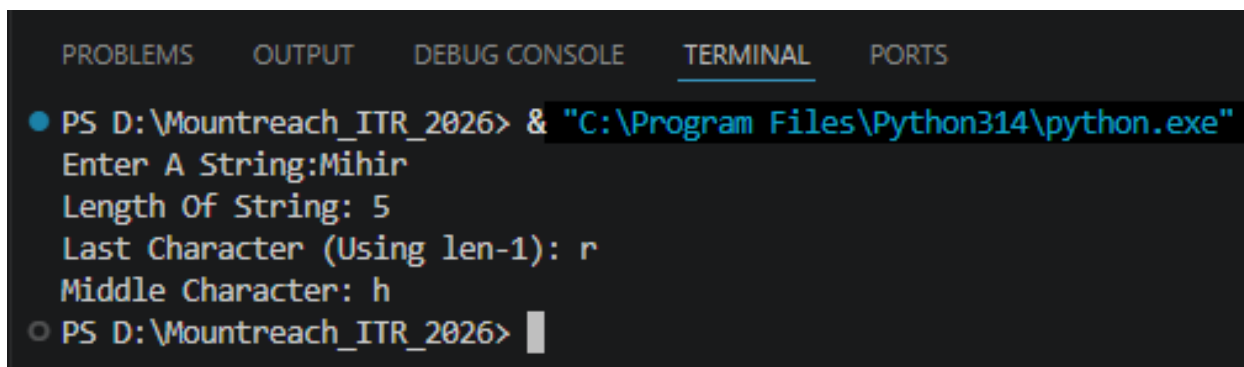
4. len() Function

Write a program that:

- Takes a string as input.
- Prints the length of the string using len().
- Prints the last character using len() (without using negative index).
- Prints the middle character if the length is odd.

Add comments on common mistakes with len().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
Length Of String: 5
Last Character (Using len-1): r
Middle Character: h
PS D:\Mountreach_ITR_2026>
```

Program Code:

```
user_string = input("Enter A String:")

# len() Returns The Total Number Of Characters In The String
string_length = len(user_string)
print("Length Of String:", string_length)

print("Last Character (Using len-1):", user_string[string_length - 1])

# Middle Character Is Only Possible When Length Is Odd
if string_length % 2 != 0:
    print("Middle Character:", user_string[string_length // 2])
else:
    print("String Has Even Length (No Middle Character)")
```

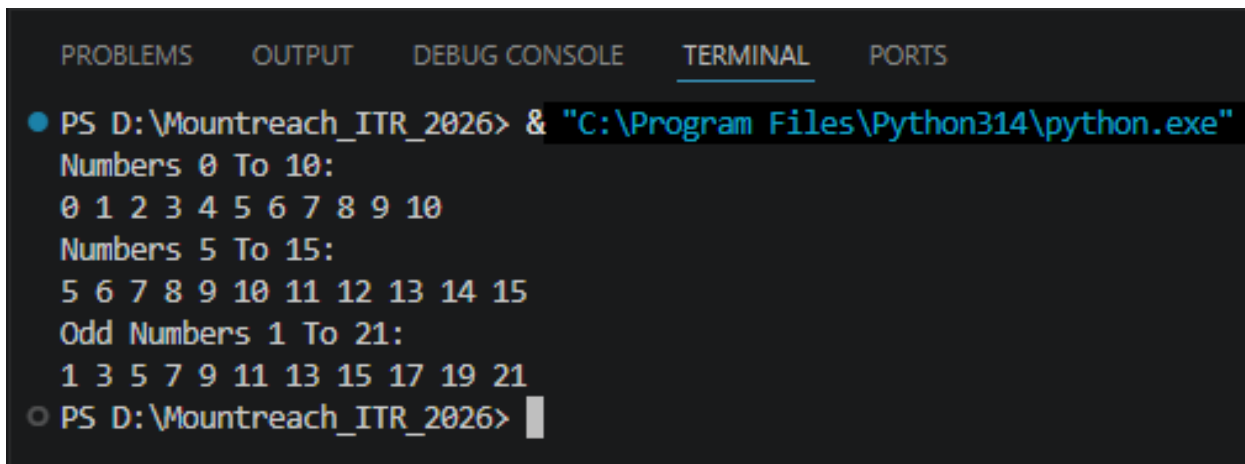
5. range() - Basic Forms

Write a program using for loop and range() to:

- Print numbers from 0 to 10.
- Print numbers from 5 to 15.
- Print odd numbers from 1 to 21.

Explain the three forms of range() in comments.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Numbers 0 To 10:
0 1 2 3 4 5 6 7 8 9 10
Numbers 5 To 15:
5 6 7 8 9 10 11 12 13 14 15
Odd Numbers 1 To 21:
1 3 5 7 9 11 13 15 17 19 21
○ PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
# Three Forms Of range():
#   range(stop)           -> 0, 1, 2, ..., stop-1
#   range(start, stop)    -> start, start+1, ..., stop-1
#   range(start, stop, step) -> start, start+step, ..., < stop

# range(stop): 0 To 10 Inclusive – Use stop=11
print("Numbers 0 To 10:")
for num in range(11):          # range(stop)
    print(num, end=" ")

# range(start, stop): 5 To 15 Inclusive – stop=16
print("\nNumbers 5 To 15:")
for num in range(5, 16):      # range(start, stop)
    print(num, end=" ")

# range(start, stop, step): Odd Numbers 1 To 21, step=2
print("\nOdd Numbers 1 To 21:")
for num in range(1, 22, 2):    # range(start, stop, step)
    print(num, end=" ")
```

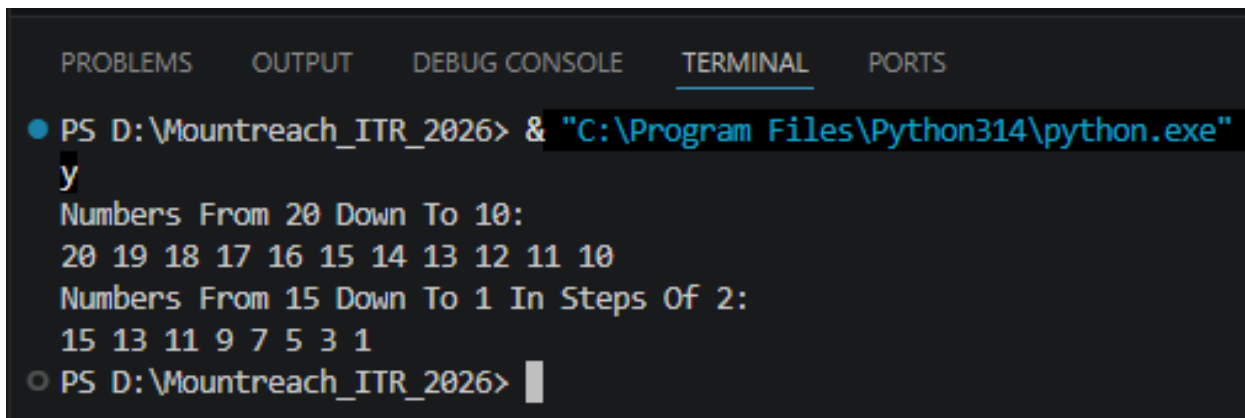
6. range() With Negative Step

Use range() to print:

- Numbers from 20 down to 10 (decreasing).
- Numbers from 15 down to 1 in steps of 2.

Print each sequence on a new line with proper messages.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
y
Numbers From 20 Down To 10:
20 19 18 17 16 15 14 13 12 11 10
Numbers From 15 Down To 1 In Steps Of 2:
15 13 11 9 7 5 3 1
PS D:\Mountreach_ITR_2026> 
```

Program Code:

```
# Negative Step Counts Down (Start Must Be Greater Than Stop)
# range(start, stop, step) Where step < 0 Goes From start Down To stop+1

print("Numbers From 20 Down To 10:")
for num in range(20, 9, -1):    # stop=9 so 10 is included
    print(num, end=" ")

print("\nNumbers From 15 Down To 1 In Steps Of 2:")
for num in range(15, 0, -2):    # stop=0 so 1 is included
    print(num, end=" ")
```

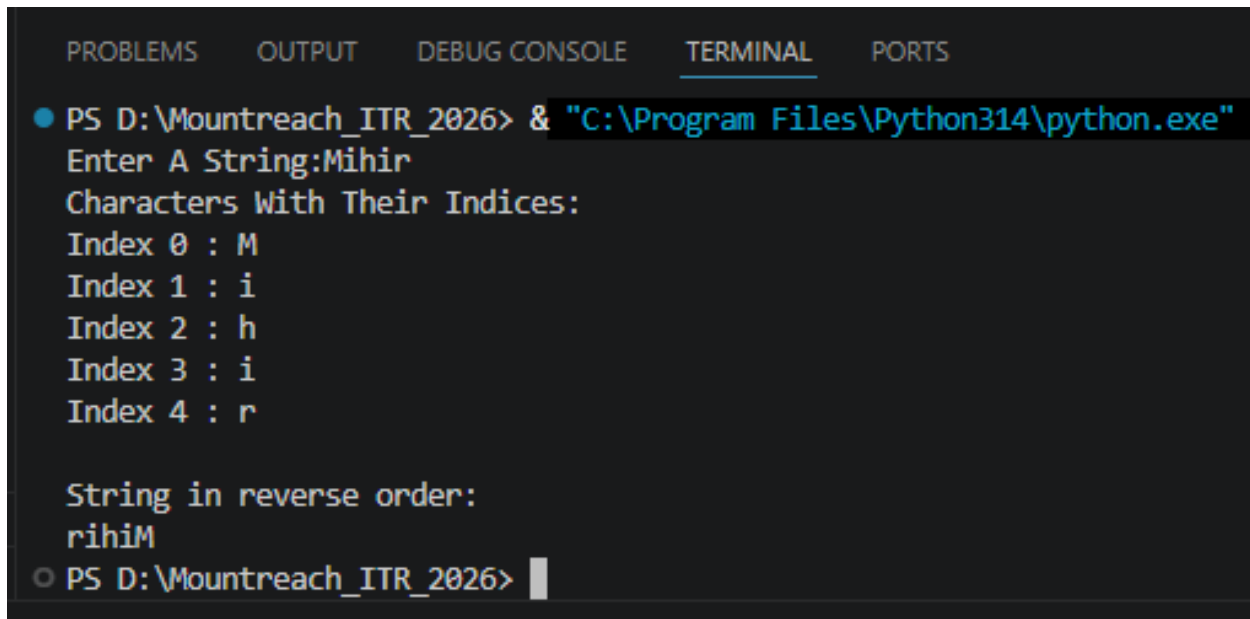

7. Combined: Strings + range()

Write a program that takes a string from the user.

Using a for loop and range() with len(), print:

- Each character of the string one by one (with its index).
- The string in reverse order using negative step in range().

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir
Characters With Their Indices:
Index 0 : M
Index 1 : i
Index 2 : h
Index 3 : i
Index 4 : r

String in reverse order:
rihiM
○ PS D:\Mountreach_ITR_2026>
```

Program Code:

```
user_string = input("Enter A String:")
str_length = len(user_string)

# Print Each Character With Its Index Using range(len(string))
print("Characters With Their Indices:")
for i in range(str_length):
    print("Index", i, ":", user_string[i])

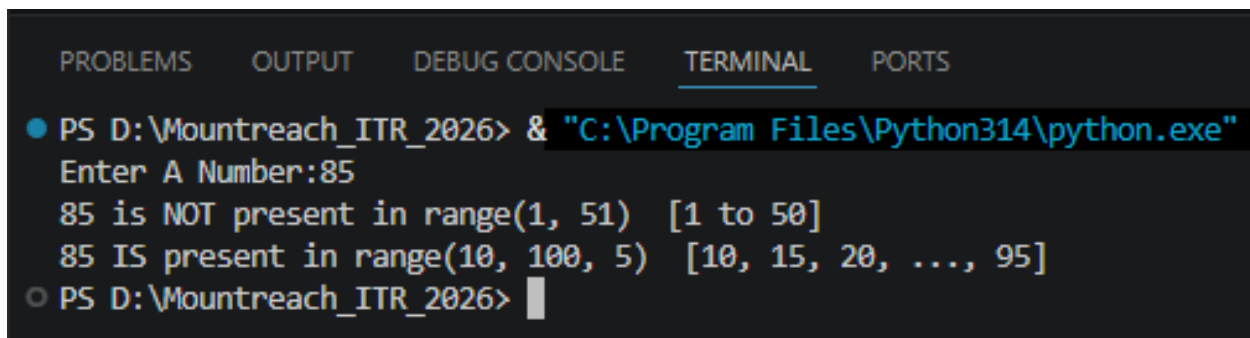
# Reverse Order: Range Starts From Last Index (len-1) Down To 0, step=-1
print("\nString in reverse order:")
for i in range(str_length - 1, -1, -1): # stop=-1 So Index 0 Is Included
    print(user_string[i], end="")
```

8. Membership With range()

Write a program that:

- Takes a number as input from the user.
- Checks if the number is present in range(1, 51) and in range(10, 100, 5).
- Prints clear messages for each check.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A Number:85
85 is NOT present in range(1, 51) [1 to 50]
85 IS present in range(10, 100, 5) [10, 15, 20, ..., 95]
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
number = int(input("Enter A Number:"))

if number in range(1, 51):
    print(number, "IS present in range(1, 51) [1 to 50]")
else:
    print(number, "is NOT present in range(1, 51) [1 to 50]")

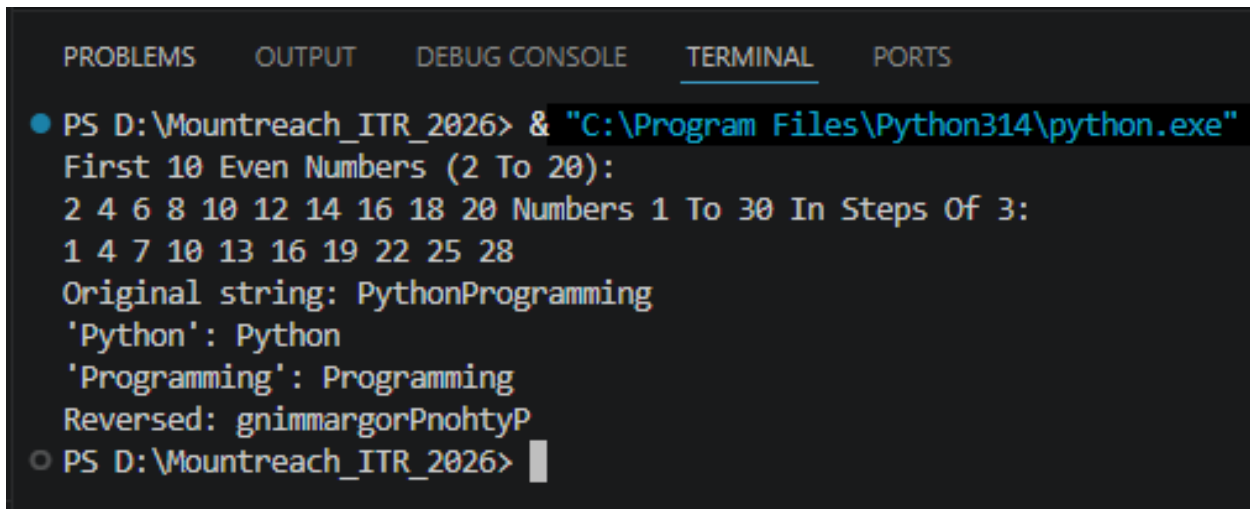
if number in range(10, 100, 5):
    print(number, "IS present in range(10, 100, 5) [10, 15, 20, ..., 95]")
else:
    print(number, "is NOT present in range(10, 100, 5) [10, 15, 20, ..., 95]")
```

9. Slicing + range()

Write a program that prints the following using range() and string slicing concepts where applicable:

- First 10 even numbers (2 to 20).
- Numbers from 1 to 30 in steps of 3.
- A string "PythonProgramming" sliced to show "Python", "Programming", and the reverse.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
First 10 Even Numbers (2 To 20):
2 4 6 8 10 12 14 16 18 20 Numbers 1 To 30 In Steps Of 3:
1 4 7 10 13 16 19 22 25 28
Original string: PythonProgramming
'Python': Python
'Programming': Programming
Reversed: gnimmargorPnohtyP
○ PS D:\Mountreach_ITR_2026> █
```

Program Code:

```
# First 10 Even Numbers: range(start=2, stop=21, step=2)
print("First 10 Even Numbers (2 To 20):")
for num in range(2, 21, 2):
    print(num, end=" ")

# Numbers 1 To 30 In Steps Of 3
print("Numbers 1 To 30 In Steps Of 3:")
for num in range(1, 31, 3):
    print(num, end=" ")

# String Slicing On "PythonProgramming"
text = "PythonProgramming"
print("\nOriginal string:",text)
print("'Python':", text[:6])           # Index 0-5
print("'Programming':", text[6:])     # Index 6 To End
print("Reversed:", text[::-1])        # step=-1
```

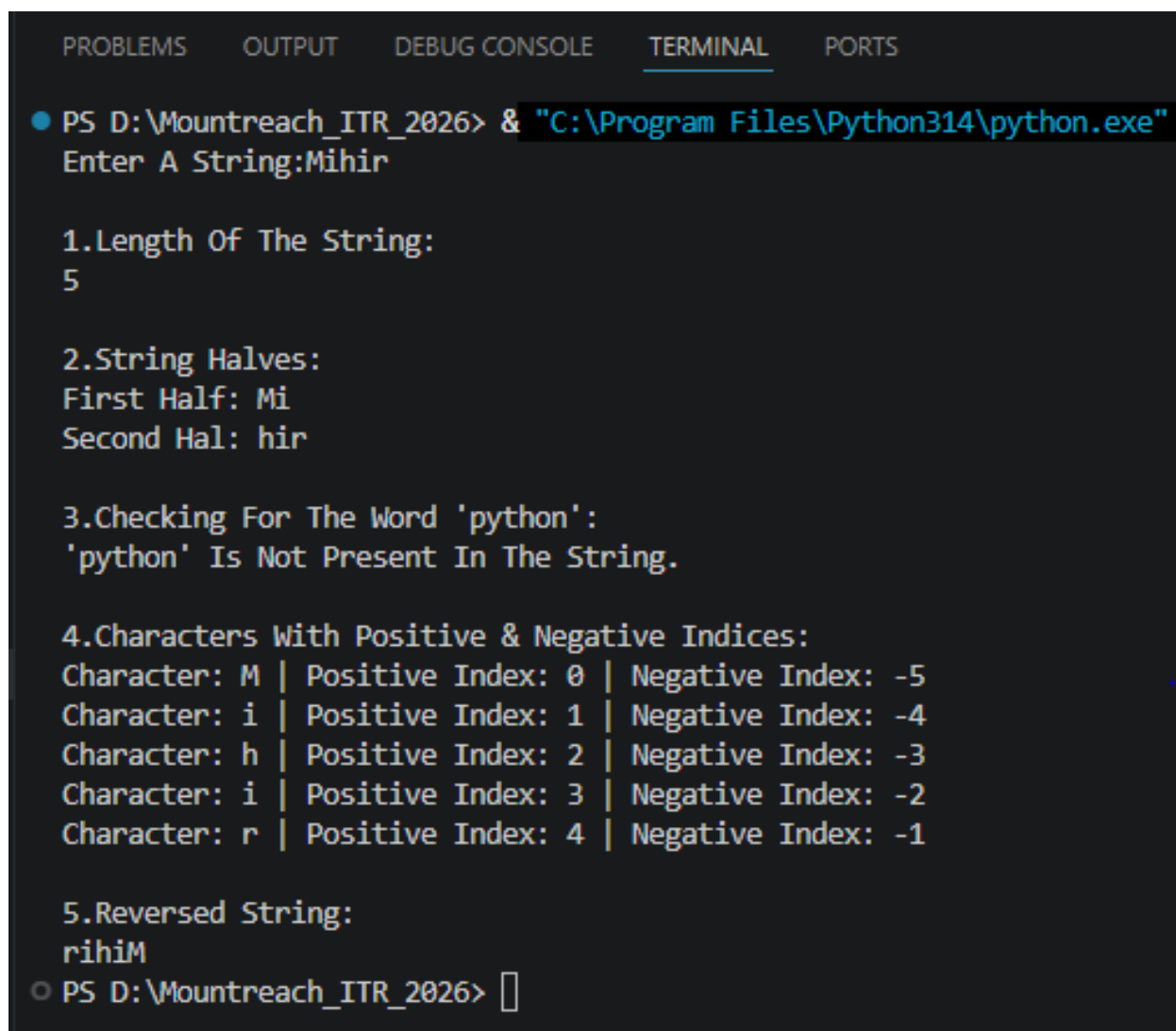
10. Mini Project - Combined

Create a **String Analyzer** program:

- Take a string input from the user.
- Using functions or directly:
 - a) Print length of the string.
 - b) Print first half and second half using slicing.
 - c) Check if the word "python" (case insensitive — you may convert) is present.
 - d) Use range() to print all characters with their positive and negative indices.
 - e) Print the string in reverse.

Add proper comments and use meaningful variable names.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach_ITR_2026> & "C:\Program Files\Python314\python.exe"
Enter A String:Mihir

1.Length Of The String:
5

2.String Halves:
First Half: Mi
Second Hal: hir

3.Checking For The Word 'python':
'python' Is Not Present In The String.

4.Characters With Positive & Negative Indices:
Character: M | Positive Index: 0 | Negative Index: -5
Character: i | Positive Index: 1 | Negative Index: -4
Character: h | Positive Index: 2 | Negative Index: -3
Character: i | Positive Index: 3 | Negative Index: -2
Character: r | Positive Index: 4 | Negative Index: -1

5.Reversed String:
rihiM
○ PS D:\Mountreach_ITR_2026> □
```

Program Code:

```
# String Analyzer Program

# Take String Input From The User
user_string = input("Enter A String:")

# Print Length Of The String
print("\n1.Length Of The String:")
print(len(user_string))

# Print First Half & Second Half Using Slicing
middle_index = len(user_string) // 2

first_half = user_string[:middle_index]
second_half = user_string[middle_index:]

print("\n2.String Halves:")
print("First Half:", first_half)
print("Second Hal:", second_half)

# Check If The Word "python" Is Present (Case-Insensitive)
print("\n3.Checking For The Word 'python':")

if "python" in user_string.lower():
    print("'python' Is Present In The String.")
else:
    print("'python' Is Not Present In The String.")

# Print All Characters With Their Positive And Negative Indices
print("\n4.Characters With Positive & Negative Indices:")

string_length = len(user_string)

for positive_index in range(string_length):
    negative_index = positive_index - string_length
    print(
        "Character:" ,user_string[positive_index],"| "
        "Positive Index:", positive_index,"| "
        "Negative Index:",negative_index
    )

# Print The String In Reverse
print("\n5.Reversed String:")
print(user_string[::-1])
```